



Fun Fact

It took the radio 38 years, & the television a short 13 years, it took the World Wide Web only 4 years to reach 50 million.



Exoplanet shining

At 50 light years away, visible light from an exoplanet, 51 Pegasi b has been detected for the first time
<http://dgit.in/ExoLight>



NETGEAR RN104

or FTP server is accomplished with just a few clicks. However, this level of simplicity does come at a price. It isn't unreasonable to expect NAS devices with flexible functionalities to be priced in the high 30s to the mid 40s.

Most devices do have low-power consuming hardware which work just fine for tasks generally seen in a SOHO environment. These are generally equipped with Celeron/Atom processors and quite a lot of them even have SoCs designed specifically for NAS devices. And almost all of them now have Gigabit Ethernet connectivity with higher models having more than one ethernet port. Add to that support for link aggregation and you're pretty much worry-free when it comes to bandwidth. The average household doesn't have multiple machines running RAID setups so hitting the bandwidth cap is a notion that most of us need not entertain. Memory matters for file transfers and helps when you are running a lot of processes on the NAS. Most manufacturers

do provide enough RAM based on what all features the NAS has to offer, and the ones that have a lot of apps to offer can be upgraded since these use SO-DIMMs for most models.

Most NAS devices come with USB ports with which you can take a one button backup. It's a convenient feature, however, if case your NAS comes with just one USB port and you need to make use of the UPS link functionality then you might have to forgo either the UPS or the backup functionality or get a USB hub. USB 3.0 ports are a much desired option and if you do get USB 3.0 you need not worry about other interfaces since the bandwidth provided by USB 3.0 is at par with what older protocols like eSATA and Firewire could provide.

If you're looking towards getting a NAS and all of the above are the same between two models and you just can't seem to make up your mind then you might want to look at the file system. Generally, you'd just see EXT4 but there are instances where you might end up with

a NAS that supports Btrfs. Pronounced as 'Butter FS', 'Better FS', 'B-Tree FS' or simply as 'Butterface', this new file system has been coded from scratch and adds features that were found to be lacking in EXT4. One of the most talked of feature is that a lot of operations can be performed on this filesystem without unmounting the drive. So increasing and decreasing volumes and defragmenting can be performed while the drive is online. Future updates even include support for encryption and online verification of the file system. If you've used FreeNAS and found ZFS to be optimal then Btrfs aims to achieve a similar feature set.

One last point that we'd like to elaborate upon is that these devices are all running on very low powered processors and are specialised for home / SOHO environments. So expecting performance akin to an enterprise file server is unrealistic. They are primarily suited for storing documents, photos and media files and the specifications of the SoCs used in these devices reflect the same. These are just some of the things that you need to bother with when deciding over a

NAS device. Let's take a look at the contenders we have for the test with regards to how well they're built.

BUILD QUALITY SYNOLOGY DS214PLAY

The Synology DS214play at first glance seems a little deceiving. It's exterior is made out of plastic but once you pick it up and have a peak inside you start to see how well it is put together. The glossy front panel is held in place by four rubber disks which lock into the crevices between the drive bays which are in turn kept in place by two locking tabs at the top. So there is no worrying about the bays coming out when you pull the cover off. The front also has an SD card slot right next a USB 2.0 port. Both can be used with the one-button backup feature. Lastly, there are LED indicators on the front. The sides are emblazoned with the Synology logo and a closer inspection reveals that the engraving have ventilation slits. Air is channelled into the device from the rear via a 90mm fan which run quite smoothly and makes no noise. Underneath the fan you find two USB



Thecus N2310